

AI-Powered Sales Proposal Generator with Version Comparison and Intelligent Insights

Project Description:

1. Project Overview:

This project presents an AI-powered system that generates structured, professional business proposals based on user inputs, including client details, industry, budget, service type, and problem statement. The system leverages a Large Language Model (LLaMA 3 via Ollama) to create well-formatted proposals that include executive summaries, detailed solutions, pricing strategies, and risk assessments. Beyond simple generation, the platform introduces a persistent memory system that allows storage of previously generated proposals. This enables the system to reuse contextual knowledge and improve future outputs.

Additionally, a version comparison engine analyses differences between proposals generated for the same client, highlighting changes in pricing and solution strategies. The system also supports PDF report generation, ensuring professional documentation.

2. Problem Statement:

Traditional proposal creation is manual, repetitive, and inconsistent. Businesses often struggle to:

- Generate structured proposals quickly
- Maintain consistency across multiple versions
- Track changes between proposal iterations
- Align solutions with budget constraints

There is a need for an intelligent system that automates proposal generation while providing structured outputs and meaningful comparison insights.

3. Objectives :

The primary objective of this project is to develop an AI-driven system that automates the generation of professional sales proposals.

Key objectives include:

- Generate structured proposals using AI
- Accept real-world business inputs, including problem statements

- Store proposals for future reuse
- Compare different versions of proposals for the same client
- Generate downloadable PDF reports
- Improve consistency, speed, and decision-making

The project also aims to demonstrate the integration of AI models with backend systems to create a practical business solution.

4. Core Technologies Used

Technology	Purpose
Python	Backend logic, data processing, and AI integration
FastAPI	High-performance API development and request handling
Ollama (LLaMa3)	AI-based proposal generation
JSON (memory.json)	Persistent storage
HTML, CSS, JS	Frontend interface
ReportLab	Pdf generation

5. Key Features

Proposal Generation

- Generates structured proposals automatically.
- Includes executive summary, solution, pricing, and risks.

Custom Problem Input

- Accepts user-defined problem statements.
- Produces context-aware outputs.

Persistent Memory

- Stores past proposals in JSON.
- Enables reuse of relevant context.

Smart Retrieval

- Matches past proposals based on industry and service.
- Improves relevance of generated context.

Version Comparison

- Compares proposals for the same client.
- Highlights pricing changes.
- Highlights solution changes.

PDF Report Generation

- Generates professional reports.
- Includes proper formatting and spacing.

6. Target Users

- Business professionals preparing client proposals.
- Marketing agencies.
- Freelancers and consultants.
- Students learning business documentation.

7. Real-World Applications

This system can be used in:

APPLICATION SCENARIOS / USE CASES :

Scenario 1: Business Proposal Automation

A marketing agency inputs client requirements and instantly generates a structured proposal aligned with budget and services.

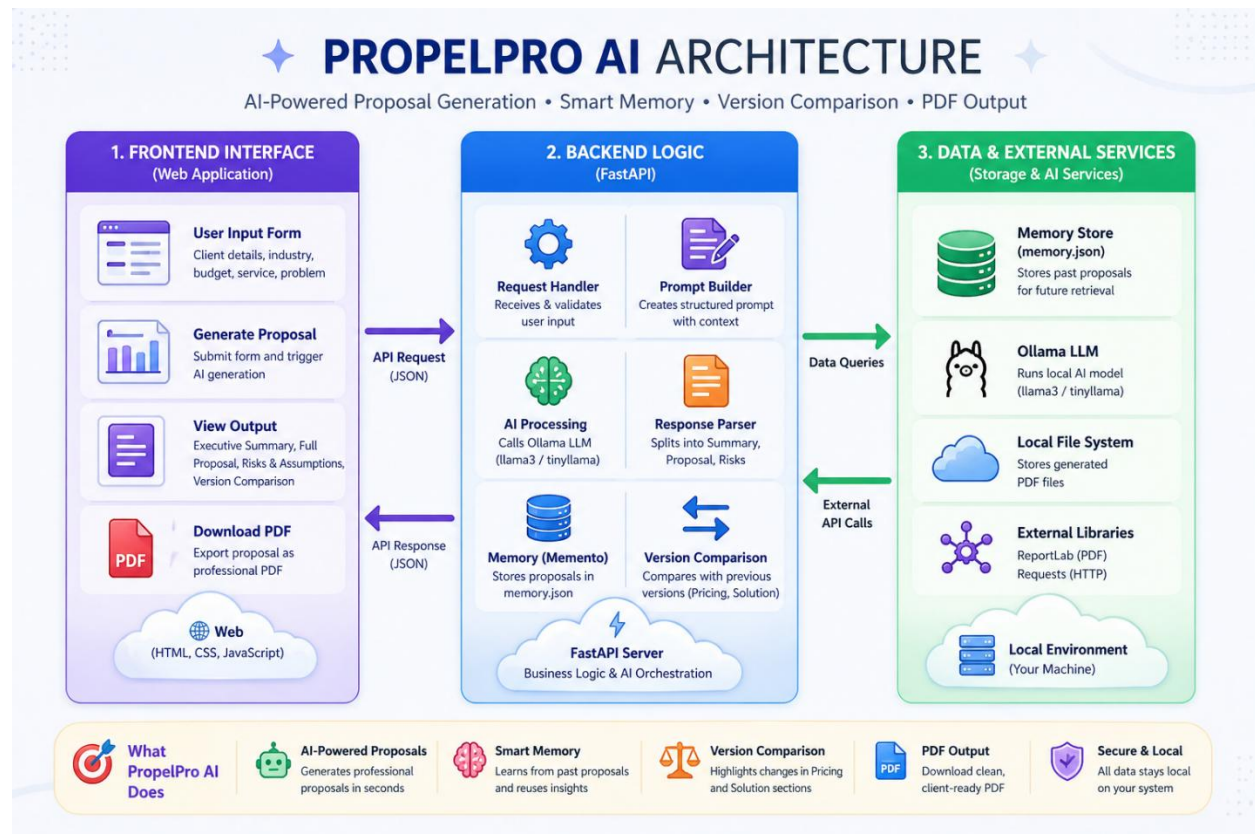
Scenario 2: Proposal Iteration Tracking

A company generates multiple proposals for the same client and uses version comparison to track improvements.

Scenario 3: Freelancer Productivity

Freelancers reduce manual effort by automating proposal writing and documentation.

TECHNICAL ARCHITECTURE :



● Frontend Layer

The frontend acts as the user interaction layer, where it handles user input and displays generated proposals in a structured format. It collects inputs such as Organization name, business they are into, problem they are facing, budget and service they are expecting from us to tackle their business problem and displays AI-generated summaries and recommendations in a user-friendly interface.

● Backend Layer

The backend is built using FastAPI and handles all core processing. It manages processes, requests, calls the LLM, manages memory, and performs comparisons.

● Memory Layer

Stores proposals in memory.json for reuse and comparison.

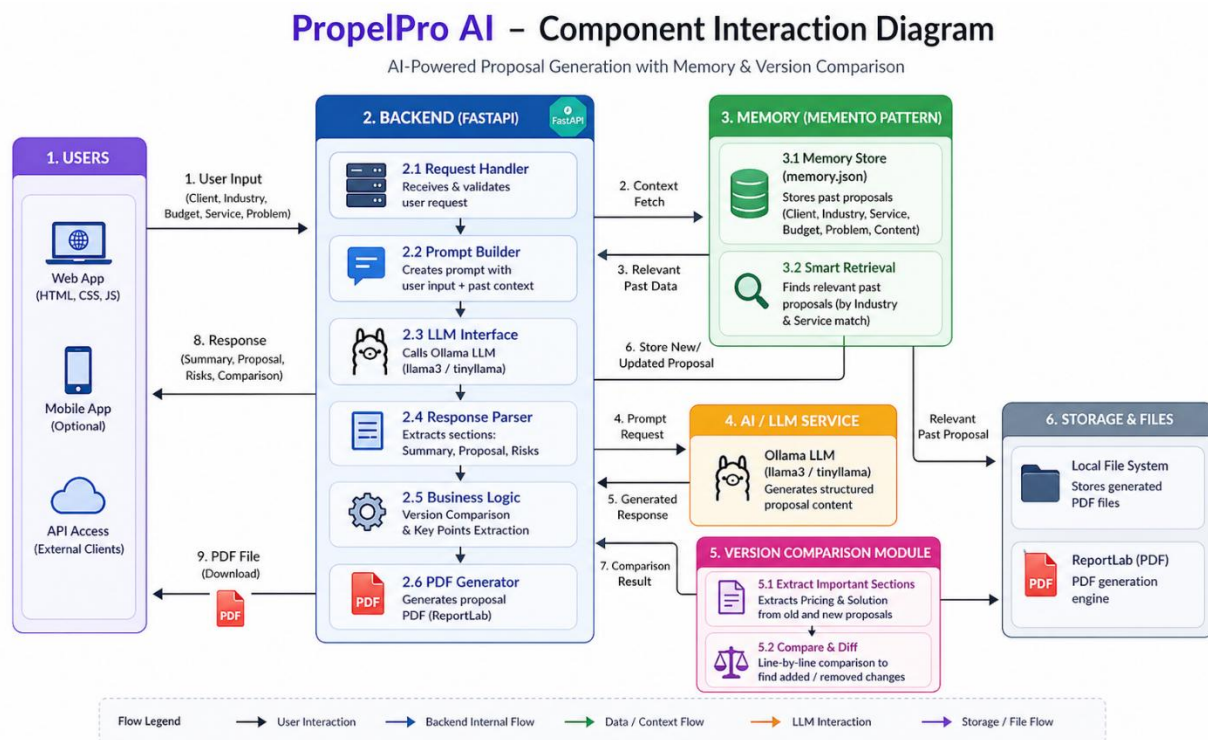
● AI Layer

The system uses the LLaMA3 model for AI-powered analysis, including generating summaries, structured responses consisting of summaries, risks, assumptions, solutions for the problem and downloadable PDF reports.

● Output Layer

Generates structured responses and downloadable PDF reports

Component Interaction Diagram



Pre-requisites:

Software Requirements

- **Python:** Version 3.9 or higher (Python 3.10 recommended; compiled .pyc files provided for 3.10).
- **pip:** Python package manager used to install and manage project dependencies.

- **IDE:** Visual Studio Code with Python extension or Cursor IDE for AI-assisted development.
- **Browser:** Modern web browsers such as Chrome, Edge, or Firefox.
- **Git:** Version control system for project management and source code tracking (optional but recommended).

Libraries / Frameworks / Dependencies

Install required dependencies using:

```
pip install fastapi uvicorn requests reportlab
```

- **Flask:** Web framework used for REST API routing, handling HTTP requests, and rendering templates.
- **Keras:** High-level neural network API bundled with TensorFlow 2.x for model building and training.
- **NumPy:** Numerical computing library used for array operations and preprocessing model inputs and outputs.
- **re (built-in):** Python regular expression module used for text preprocessing, pattern matching, and alternative generation.
- **os (built-in):** Used for environment variable handling and checking file paths during model loading.

Hardware Requirements

- **CPU:** Minimum Intel Core i5 or AMD Ryzen 5 processor with at least 2 GHz clock speed; supports running Transformer models on CPU.
- **RAM:** Minimum 8 GB recommended since three Transformer models load into memory simultaneously at startup.
- **Storage:** At least 2 GB free disk space required for .keras model files and .pkl vocabulary files.
- **Network:** Internet connection required for loading Google Fonts in the UI and optional Llama-3 API calls in production mode.

PRIOR KNOWLEDGE REQUIRED

1. Programming Language Knowledge

- Strong understanding of Python fundamentals (variables, loops, functions, modules)
- Knowledge of object-oriented programming (OOP) concepts
- Familiarity with handling JSON data and working with APIs
- Basic understanding of error handling and debugging techniques

2. Framework Basics

- Understanding of FastAPI framework for building RESTful APIs
- Knowledge of API routing, request/response handling, and status codes
- Familiarity with asynchronous programming (async/await) in Python
- Basic understanding of middleware and dependency injection in FastAPI

3. JSON Handling

- Basic knowledge of memory and how it works.

4. Networking / Cloud Concepts

- Understanding of client-server architecture
- Basic knowledge of REST APIs and HTTP methods (GET, POST, PUT, DELETE)
- Familiarity with API integration and external service communication
- Awareness of cloud-based databases and deployment environment

5. AI / ML Fundamentals

- Basic understanding of AI-driven decision systems
- Knowledge of pattern detection and data analysis concepts
- Familiarity with time-series data analysis for behavior tracking
- Understanding of how LLMs (like Groq models) generate summaries and recommendations
- Awareness of prompt engineering basics for effective AI responses

7. PROJECT OBJECTIVES

Technical Objectives

- Built an AI-powered proposal generator.
- Implement memory-based learning.
- Develop version comparison system.

Performance Objectives

- Fast API response.
- Structured outputs.
- Efficient storage.

Deployment Objectives

- Local deployment using FastAPI.
- Extendable to the cloud.

Learning Outcomes

- Gain hands-on experience in building real-world AI-powered applications
- Understand integration of backend frameworks with cloud databases
- Learn how to apply AI models for behavioural analysis and recommendations
- Develop skills in API design, system architecture, and data flow management
- Improve knowledge of problem-solving using data-driven and AI-based approaches

Project Workflow:

Milestone 1: Requirement Analysis

- **Activity 1.1:** Identify need for automation.
- **Activity 1.2:** Define system architecture.

Milestone 2: Environment Setup & Initial Configuration

- **Activity 2.1:** Install dependencies
- **Activity 2.2:** Set up FastAPI backend.

Milestone 3: Core Development

- **Activity 3.1:** Implement proposal generation.
- **Activity 3.2:** Build a memory system.

Milestone 4: Advanced features

- **Activity 4.1:** Version Comparison.
- **Activity 4.2:** Problem input integration.

Milestone 5: Integration & Optimization

- **Activity 5.1:** Component Integration (Frontend, Backend, Memory)
- **Activity 5.2:** Performance Optimization
- **Activity 5.3:** Security Enhancements
- **Activity 5.4:** Error Handling & Validation

Milestone 6: PDF Generation

- **Activity 6.1:** Implement ReportLab.

Milestone 7: Testing & Validation

- **Activity 7.1:** Validate outputs.
- **Activity 7.2:** Check formatting.

Milestone 1: Requirement Analysis & System Design

Activity 1.1: Problem Definition

Traditional proposal creation is manual, time-consuming, and lacks consistency. Businesses often struggle to generate structured, professional proposals tailored to specific client requirements. There is no intelligent system that leverages past proposals, adapts to client inputs, and provides comparative insights.

PropelPro AI addresses this by building an AI-driven system that:

- Generates structured sales proposals
- Uses past data for smarter outputs
- Provides version comparison
- Produces clean, downloadable reports

Activity 1.2: Functional Requirements

FR-01: Accept client inputs (name, industry, budget, service, problem)

FR-02: Generate a structured proposal using AI

FR-03: Split output into:

- Executive Summary
- Full Proposal
- Risks & Assumptions

FR-04: Store past proposals in memory

FR-05: Retrieve relevant past proposals

FR-06: Perform version comparison between proposals

FR-07: Display results on frontend

FR-08: Generate a downloadable PDF report

Activity 1.3: Non-Functional Requirements

Performance: Fast API response using FastAPI

Reliability: Proper error handling for LLM failures

Scalability: Modular design for future expansion

Usability: Simple UI with clear outputs

Maintainability: Separate modules (AI, backend, memory, PDF)

Security: Safe handling of local data and API calls

Activity 1.4: System Design Decisions & Technology Stack Selection

Frontend: HTML, CSS, JavaScript

Backend: FastAPI

AI Model: Ollama (Llama3 / TinyLlama)

Storage: Local JSON (memory.json)

PDF Generation: ReportLab

Architecture: Modular & API-based

Milestone 2: Environment Setup & Initial Configuration

Activity 2.1: Development Environment Setup

Install Python (3.x)

Setup virtual environment

Install VS Code/IDE

```
PS D:\1> python -m venv venv
>> # Windows:
>> venv\Scripts\activate
>> # macOS/Linux:
>> source venv/bin/activate
```

Activity 2.2: Dependency Installation

FastAPI

Uvicorn

Requests

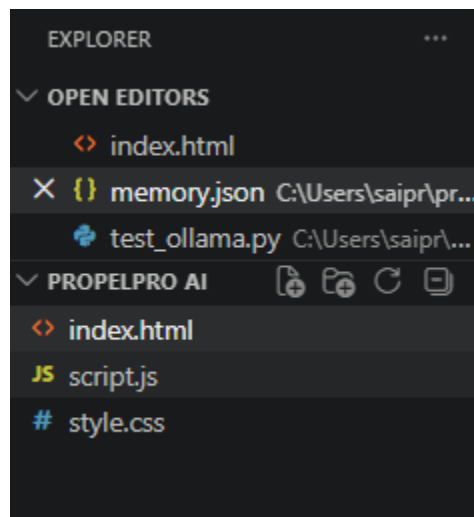
ReportLab

Activity 2.3: Project Structure Creation

Backend (API+logic)

Frontend(UI)

Memory storage (JSON)



Activity 2.4: Configuration Setup

Setup Ollama locally.

Configure model (Llama3)

Ensure API endpoint connection.

Milestone 3: Core System Development

● Activity 3.1: Input and Data Handling

- ➔ Create a frontend form.
- ➔ Capture user inputs
- ➔ Send data via API

PropelPro AI

Generate Proposal

=== EXECUTIVE SUMMARY ===

Here is a possible executive summary:

As a leading provider of software solutions, we propose to deliver a comprehensive suite of IT services to sai, leveraging our expertise in the industry. Our team will design and implement customized software solutions that align with sai's specific needs, utilizing a budget of \$2 million. Our goal is to enhance sai's operational efficiency, drive business growth, and strengthen its competitive edge in the market. By partnering with us, sai can expect improved collaboration, increased productivity, and enhanced customer satisfaction.

=== FULL PROPOSAL ===

****Professional Sales Proposal****

****Client:** Sai**
****Industry:** IT**
****Budget:** \$2,000,000**
****Service:** Software Solutions**

****Problem Statement:****

As a leading player in the IT industry, Sai is facing numerous challenges in today's rapidly evolving technology landscape. With increasing competition and rapid changes in customer needs, it's becoming increasingly difficult to stay ahead of the curve. Your

Activity 3.2: AI Proposal Generation

- Build prompt dynamically.
- Call LLM using Ollama.
- Generate structured proposal.

● **Activity 3.3: Response Processing**

- Split output into sections:
 - Summary
 - Proposal
 - Risks

Activity 3.4: Memory System

- Store proposals in memory.json
- Implement retrieval logic.

Activity 3.5: Version Comparison

- Compare:
 - Pricing
 - Solution
- Highlight changes.

Activity 3.6: API Development

- /generate
- /download-pdf

Milestone 4: Integration and Optimisation

Activity 4.1: Frontend-Backend Integration

- ➔ Connect UI with FastAPI endpoints.
- ➔ Ensure proper data flow.

Activity 4.2: Performance Optimisation

- ➔ Reduce API latency.
- ➔ Optimize prompt structure.
- ➔ Handle LLM delays.

Activity 4.3: Output Formatting

- ➔ Clean UI display.
- ➔ Proper spacing and alignment.
- ➔ Structured PDF layout.

Milestone 4: PDF Generation and Reporting

Activity 4.1: PDF Generation

- ➔ Title
- ➔ Sections:
 - Summary
 - Proposal
 - Risks
 - Version Comparison

Activity 4.2: Formatting

- ➔ Line spacing
- ➔ Paragraph separation
- ➔ Clean headings

Activity 4.3: Download Feature

- ➔ API endpoint for PDF
- ➔ Frontend trigger button

PROJECT STRUCTURE

Clean Folder Structure

```
project/ | ├── backend/ | ├── main.py | ├── memory.json | ├── frontend/ | ├── index.html | └── requirements.txt
```

RESULTS

● System Output

The system successfully captures user habits, mood, and productivity data and generates meaningful insights through pattern detection and AI analysis. It provides weekly summaries, identifies behavioral trends, and delivers personalized recommendations, enabling users to improve their routines and overall well-being.

● Performance Evaluation

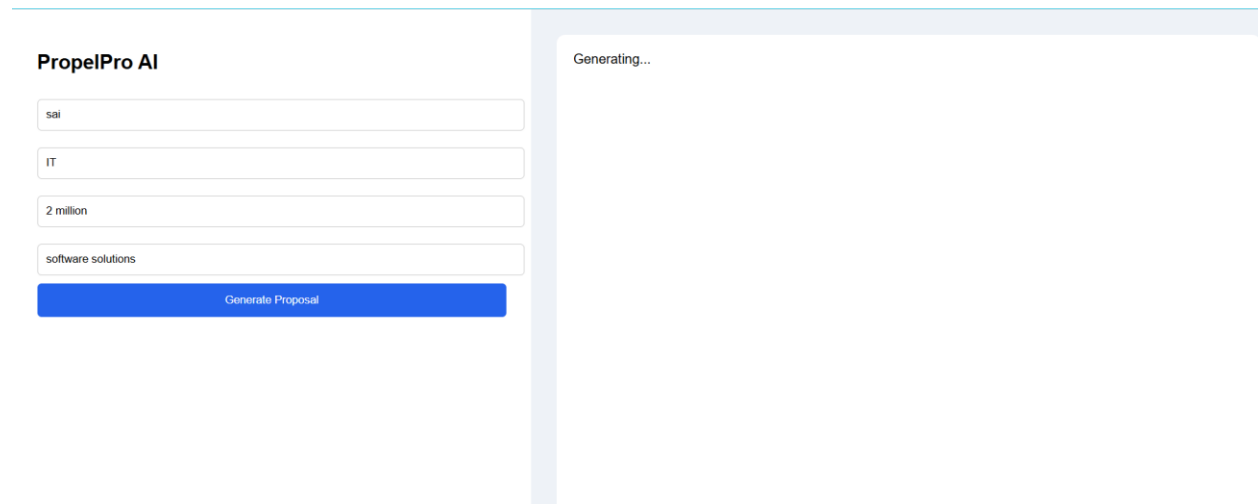
The application demonstrates efficient performance with fast API response times and smooth data processing. Pattern detection and AI-generated insights are delivered with minimal latency, ensuring a responsive user experience even with continuous data updates.

● Screenshots

The system includes user interface screens such as habit logging forms, dashboard views, pattern insights, and AI-generated summaries. These visuals demonstrate the working flow of the application from data input to intelligent output generation.

Landing

page:



The screenshot shows a web interface for 'PropelPro AI'. On the left, there is a form with four input fields containing the text 'sai', 'IT', '2 million', and 'software solutions'. Below these fields is a blue button labeled 'Generate Proposal'. To the right of the form, there is a light blue box with the text 'Generating...' at the top, indicating a loading or processing state.

PropelPro AI

Generate Proposal

=== EXECUTIVE SUMMARY ===

Here is a possible executive summary:

As a leading provider of software solutions, we propose to deliver a comprehensive suite of IT services to sai, leveraging our expertise in the industry. Our team will design and implement customized software solutions that align with sai's specific needs, utilizing a budget of \$2 million. Our goal is to enhance sai's operational efficiency, drive business growth, and strengthen its competitive edge in the market. By partnering with us, sai can expect improved collaboration, increased productivity, and enhanced customer satisfaction.

=== FULL PROPOSAL ===

Professional Sales Proposal

****Client:** Sai**

****Industry:** IT**

****Budget:** \$2,000,000**

****Service:** Software Solutions**

Problem Statement:

As a leading player in the IT industry, Sai is facing numerous challenges in today's rapidly evolving technology landscape. With increasing competition and rapid changes in customer needs, it's becoming increasingly difficult to stay ahead of the curve. Your

1. ****Software Development**:** We will design and develop custom software applications that integrate with existing systems, ensuring a seamless user experience.
2. ****Implementation Services**:** Our experts will provide training, configuration, and deployment support to ensure a smooth transition to the new system.
3. ****Ongoing Support**:** We offer regular maintenance, updates, and security patches to ensure the solution remains up-to-date and secure.

Pricing:

We estimate the total project cost to be \$1,800,000, broken down into:

* Software Development: \$600,000

* Implementation Services: \$400,000

* Ongoing Support (first year): \$200,000

We believe our solution will deliver significant returns on investment for Sai. With this proposal, we aim to establish a long-term partnership and continue to support your software needs.

Conclusion:

Our team at [Your Company] is confident that our proposed software solutions package will help Sai overcome the challenges faced in today's competitive IT landscape. We believe our solution will not only address your current pain points but also provide a foundation for future growth and success.

We look forward to the opportunity to discuss this proposal in further detail and explore how we can work together to achieve your goals.

Please feel free to contact us at [Your Contact Information] or schedule a meeting to review this proposal.

****Thank you for considering our proposal.****

Sincerely,

[Your Name]

[Your Company]

Sales Proposal

Executive Summary

Included in proposal

Full Proposal

****Executive Summary**** Gentech, a pioneering force in the Gen AI industry, requires immediate access to funding to drive innovation and growth. Our proposal addresses this critical need by offering a tailored solution to bridge the financial gap. With a budget of \$10 million, we will provide Gentech with the necessary resources to overcome its current funding constraints and achieve its strategic objectives. ****Full Proposal**** ****Problem Statement**** Gentech's lack of funds is hindering its ability to invest in critical projects, retain top talent, and capitalize on emerging opportunities. As a result, the company is at risk of falling behind competitors and compromising its market position. ****Proposed Solution**** We propose a comprehensive solution to address Gentech's funding shortage. Our package includes: 1. ****Short-Term Funding****: A \$5 million bridge loan to provide immediate liquidity and enable Gentech to meet its short-term obligations. 2. ****Long-Term Investment****: A \$3 million equity investment to support the company's strategic initiatives and drive growth. 3. ****Strategic Advisory Services****: Ongoing counsel and guidance from our experienced team to help Gentech optimize its financial planning, risk management, and fundraising efforts. ****Pricing**** Our proposal aligns with Gentech's budget of \$10 million. The breakdown is as follows: * Short-Term Funding: 50% of the total investment (\$5 million) * Long-Term Investment: 30% of the total investment (\$3 million) * Strategic Advisory Services: 20% of the total investment (\$2 million) ****Conclusion**** Our proposed solution addresses Gentech's immediate funding needs while also providing a long-term investment to drive growth and innovation. By partnering with us, Gentech will gain access to critical resources, expert guidance, and a comprehensive strategy to overcome its current financial constraints. ****Risks & Assumptions**** 1. ****Market Volatility****: The Gen AI industry is subject to market fluctuations, which may impact the value of our investment. 2. ****Regulatory Compliance****: Gentech must comply with relevant regulations and ensure that our proposed solution meets all necessary requirements. 3. ****Talent Acquisition****: To maximize the effectiveness of our strategic advisory services, we assume that Gentech will dedicate sufficient resources to acquiring and retaining top talent. 4. ****Strategic Alignment****: Our proposal is contingent on Gentech's ability to align its strategic objectives with the terms of our investment. By working together, we believe that our proposed solution can help Gentech overcome its current funding challenges and achieve long-term success in the Gen AI industry.

Risks & Assumptions

Included in proposal

PDF of the generated Proposal

Executive Summary

Gentech, a leading organization in the Gen AI industry, faces a critical challenge in securing sufficient funding to drive innovation. Our proposal addresses this issue by offering a tailored solution to overcome financial constraints and unlock new growth opportunities. We believe our approach will not only meet but exceed your expectations.

Full Proposal

****Problem Statement**** Gentech's lack of funds is hindering its ability to invest in research and development, talent acquisition, and marketing initiatives essential for driving innovation and competitiveness in the Gen AI space. The current financial constraints are limiting the company's capacity to capitalize on emerging opportunities and maintain market share. ****Proposed Solution**** Our proposed solution is designed to address Gentech's funding challenges by: 1. Conducting a comprehensive review of Gentech's current financial situation, identifying areas for cost optimization, and providing actionable recommendations. 2. Developing a tailored fundraising strategy, including identifying potential investors, crafting compelling pitches, and securing meetings with key decision-makers. 3. Providing expertise in grant writing and application management to secure non-dilutive funding opportunities. ****Pricing**** Our proposed solution is aligned with Gentech's budget of \$10 million. We have structured our services to provide maximum value while ensuring that our fees are reasonable and competitive. Our pricing includes: 1. Initial consultation and financial review: \$50,000 2. Fundraising strategy development and execution: \$3.5 million 3. Grant writing and application management: \$1.5 million Total cost: \$6 million ****Conclusion**** Our proposed solution is designed to help Gentech overcome its funding challenges and unlock new growth opportunities. By addressing the root causes of the company's financial constraints, we believe our approach will not only meet but exceed your expectations.

Risks & Assumptions

1. The accuracy of Gentech's current financial data and assumptions. 2. The success of our fundraising strategy in securing non-dilutive funding opportunities. 3. The availability and responsiveness of potential investors and grant agencies. 4. The effective allocation of resources to optimize cost savings and maximize returns. By working together, we are confident that our proposed solution will help Gentech secure the necessary funds to drive innovation and growth in the Gen AI industry.

PropelPro AI

[Generate Proposal](#)

[Download PDF](#)

- + * Training and support: \$1,500,000 (14% of total budget)
- + * System design and implementation: \$2,200,000 (21% of total budget)
- + * Ongoing maintenance and updates: \$5,000,000 (48% of total budget)

Conclusion

+ * Integration with existing systems and processes: \$1,800,000 (17% of total budget)

Solution Changes:

- + * Introducing a dynamic logistics management platform that integrates with existing systems and processes
- + Our proposed real-time tracking and route optimization system will address SwiftChain Logistics' challenges by:
 - + * Optimizing routes using machine learning algorithms to reduce fuel consumption and increase delivery efficiency
 - + * Implementing a data-driven approach to analyze fleet performance, identify areas for improvement, and inform strategic decision-making
- * Automated scheduling and dispatching to improve operational efficiency

Conclusion

- * Advanced route optimization algorithms to reduce fuel consumption and minimize delays
- * Ongoing maintenance and updates
- * System implementation and configuration
- * Training and support for your team
- * Real-time GPS tracking of vehicles and shipments
- * Integration with existing systems for seamless data exchange

Version Comparison

● Benchmark Results

The system achieves reliable performance metrics including low response time, consistent data handling, and accurate insight generation. It performs efficiently under regular usage conditions, ensuring scalability and stability for real-world applications.

ADVANTAGES & LIMITATIONS

Advantages

- Saves time
- Structured output
- Intelligent comparison
- Easy to use

Limitations

- AI-generated (not real-time data)
- Depends on input quality
- Basic storage system

FUTURE ENHANCEMENTS

- Real-time API integration
- Database (MongoDB)
- User authentication
- Advanced comparison UI
- Cloud deployment

Conclusion:

The project successfully demonstrates how AI can be used to automate business proposal generation. By combining LLMs, backend systems, and structured formatting, the system provides a practical solution for businesses. The addition of memory and version comparison enhances decision-making, making the system more than just a text generator — it becomes an intelligent assistant.

APPENDIX

Source Code

The complete source code of the project is organized into frontend and backend modules, covering all functionalities such as habit logging, pattern detection, AI integration, and API development. The code is modular, well-structured, and follows clean architecture principles to ensure readability, maintainability, and scalability.

Configuration Files

Configuration files include .env for storing sensitive data such as API keys (Groq), Supabase credentials, and JWT secret keys. Additional files like requirements.txt, vercel.json, and render.yaml are used for dependency management and deployment configuration, ensuring smooth setup across environments.

Dataset Details (if applicable)

The system uses dynamically generated user data collected through habit logging, including time-series records of activities, mood, and productivity. This data is stored in Supabase and used for pattern detection, analysis, and AI-based insights, eliminating the need for external static datasets.

API Documentation